# 9 Designing anthropological impact: How case-based teaching makes a difference

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## Introduction

"Thank you for your attention. We hope that our insights can be useful in your future work with waste management in the municipality". With these words, anthropology student Halfdan ends a 12-minute presentation for a group of employees from the Waste Unit in the Municipality of Copenhagen. Together with 23 other students, Halfdan has in the past 12 weeks worked with the theme of "circular economies" in the public waste sector in Denmark. More specifically, with his fellow students, Johanne, Siri, and Ellen, he has explored the theme with the concept of "careful littering", which the group has developed in order to reframe waste handling as an act of care for the environment and, hence, a pleasurable activity. Although aiming at sharing applicable insights and perspectives with municipal employees, the students deliberately did not provide clear and instrumental solutions to the questions and needs raised by the municipality. Rather, they sought to explore the issue of waste handling in an empirical and critical way that essentially allowed them to deconstruct and reconfigure prevailing viewpoints and practices – and, thus, to open up to new ways of thinking, doing, and being. "[T]o change our understanding", as Gibson-Graham (2008, p. 615) has phrased it, "*is* to change the world, in small and sometimes major ways".

In this chapter, we describe how university-industry collaborations may unfold through case-based teaching. Drawing on our engagement in the above-mentioned course on applied anthropology on the bachelor's programme in anthropology at the University of Copenhagen, as well as our ten years of experience with external collaboration and real-life cases in teaching, we provide a hands-on description of how to organise applied or practice-oriented courses in anthropology and related disciplines. On this basis, three interconnected arguments are put forward. First, case-based teaching organised around challenges posed by actors outside the university train the students in critical thinking and constructive application as two sides of the same coin. Second, by so doing, the students provide value to the collaborating actors and the wider society by challenging established

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perspectives and practices by way of proposing novel ways of thinking, doing, and being. Third, and more broadly, this has particular subject effects as this form of teaching promotes a critical yet engaged kind of citizenship among the students.

To make these arguments, we begin by outlining the broader context of recent science policies that serve as a significant backdrop to our case-based teaching course. Then, we present our distinct approach to the course which draws much inspiration from design anthropology and the notion of ontological design. In the following section, we describe the organisation of the course in detail, emphasising the specific format, activities, and progression. This leads us to present the above-mentioned student project in more detail as an example of a case-based learning process that also contributed productively to the work of our collaborating partner, the Waste Unit. On this basis, we discuss the tensions and challenges involved in case-based teaching, including the anthropological impact made by the students. Finally, we conclude by linking our argument to the broader issue of humanities meeting business – or, in this context, students of anthropology meeting public organisations – in terms of the kinds of subject or citizenship being promoted by our approach to case-based teaching.

## The entrepreneurial university and mission-driven teaching

In the past decades, European public universities have been subject to substantial and wide-ranging science policies. As Gibbons et al. (1994) emphasised in the 1990s, public universities have undergone a transition from mainly producing scientific knowledge within the echelons of the academic institution – what they term "Mode 1" – towards open collaborations beyond disciplinary and academic boundaries – what they term "Mode 2". By closely interacting with diverse actors from outside academia, not least commercially driven ones, public universities are in Mode 2 envisioned to produce "robust knowledge" of societal relevance (see also Nowotny, Scott, and Gibbons, 2001, p. 166). Although developed as a critique of this transition from Mode 1 to Mode 2, the Triple Helix policy model similarly expresses a transformation of public universities (Etzkowitz, 2008). The model envisions researchers and students who collaborate with external actors for the purpose of turning frontier knowledge into innovative solutions (Etzkowitz and Leydesdorff, 1997). With a systemic approach, the Triple Helix defines interactions between industry, university, and government as essential means through which to reach regional and national economic growth (Ranga and Etzkowitz, 2013). In this way, universities have expanded their dual responsibility of producing research and education with a "third mission", expected to bring about cross-boundary collaborations and innovative results (Loi and Guardo, 2015; Zomer and Benneworth, 2011).

Particularly, the term "mission" has recently gained much traction in science policies. In the wake of influential publications such as Mazzucato's *Mission Economy* (2021) and *Mission-Oriented Research & Innovation in the European Union* (2018), recent science policies support mission-oriented projects, which, according to Mazzucato, provide an opportunity, approach, and solution to both current and future societal challenges. The objective is to design missions of frontier knowledge production that enable bottom-up experimentation and systemwide innovation (2018, p. 4). Similar to the science models mentioned above, Mazzucato emphasises the importance of cross-boundary interactions between diverse actors and disciplines:

> The objective should be addressed by multiple actors, stimulating cross-discipline academic work, with a strong focus on the intersection between natural sciences, formal sciences, social sciences and humanities; collaborations across different industries; and new forms of partnerships between the public sector, the private sector and civil society organisations. (2018, p. 12)

Our own university is a case in point. During the past ten years, the management at the University of Copenhagen has initiated several strategies to enhance interdisciplinary research, external collaboration, innovative activities, and societal impact. Innovation consultancy units, centres for applied science, cross-disciplinary impact projects, hubs for student start-ups, and, recently, an innovation centre serving all faculties have been established. Furthermore, tech transfer officers have been hired to ensure intellectual property rights and licenses of new research discoveries, while innovation consultants have joined the university to facilitate and advance an entrepreneurial mind-set among researchers and students. During the past year alone, a new Green Solutions Centre has been set up to play a key role in shaping a sustainable future, and a new public policy centre has been created to solve societal problems through research and partnerships. Both of these recent initiatives are essentially borne out of, and indeed expected to feed into, the impact and mission-oriented agenda outlined above. University of Copenhagen thus pursues various strategies to strengthen its ties to the wider society and prove its value; most clearly epitomised, perhaps, in its current catchphrase on posters around the city: *til gavn siden 1479*, meaning "impact since 1479".

The effects of recent science policies on the development of public universities have not gone unnoticed, of course, by the scholars within them. Wright and Shore (2017), for instance, ask the critical question of whether the development of higher education, with its emphasis on commercialisation, entrepreneurship, efficiency, usefulness, and related ideas, will ultimately entail the death of the public university. As a powerful metaphor for the state of public universities, they draw on the term

"zombification", developed by Whelan, Walker, and Moore (2013), to stress how "the public university today exists in a state of chronic fragility, servitude and uncertainty that has left it if not 'dead', then permanently moribund and drained of autonomy and agency" (Shore and Wright, 2017, p. 18). In a related discussion, Krause-Jensen and Garsten (2014) likewise emphasise how universities are increasingly organised as private companies, focusing on "research that can be converted into a revenue stream" (p. 1). While this, according to them, may lead to institutional uniformity and harmonisation (p. 2), it also affects higher education in terms of promoting novel modes of learning (pp. 2–3).

Although such critical discussions of the commodification and corporatisation of public universities are indeed imperative, our intention in this chapter is not to assess the recent development of public universities. However, the critical reflections are important, not only because our case-based teaching is evidently embedded in this development but also because the critique has informed how we have aimed at maintaining traditional academic virtues – such as rigorous analysis, conceptual clarity, and critical reflection – while also pursuing open collaboration, mission-oriented agendas, and societal impact. Our intention, in other words, is to demonstrate how contemporary science policies that stress applicability, impact, missions, and collaboration may be turned into practice in university teaching. As such, we seek to contribute to what may be an emergent field of case-based teaching in the humanities by drawing, in particular, on the field of design anthropology, including a reorientation of the design concept. We turn to this now.

## Design anthropology and ontological design

We are, of course, not alone in having developed case-based teaching in the humanities. While the use of business cases and case competitions is widely recognised as valuable educational activities in business schools around the world (see, e.g., Burke, Carter, and Hughey 2013), as well as problem-based learning, has long constituted an established pedagogical practice (see, e.g., Moallem, Hung, and Dabbagh, 2019), the integration of real-life challenges into the teaching of anthropology and cognate disciplines has been much slower. In the Danish context, however, the past decade has seen a number of new educational initiatives introduced with a special emphasis on stakeholder collaboration, applied cultural analysis, and interdisciplinary approaches such as techno anthropology and design anthropology (see, e.g., Jensen, 2013; Jespersen et al., 2012; Halse, 2013). This development has clearly gained traction in the wake of distinct political agendas, not least a strong focus in the 2000s on users and user research in innovation projects (Damsholt and Jespersen, 2014), but it also relates to a broader dialogue between design studies and human and social sciences that has emerged in the past decade.

A significant outcome of this dialogue is the emergence of a distinct field of design anthropology (Gunn and Donovan, 2012; Gunn, Otto, and Smith, 2013; Smith et al., 2016). While anthropology has traditionally been focused on description and analysis, based on ethnographic studies of the present and, to some extent, the recent past, design has essentially been interventionist and future-oriented, "aimed at changing existing situations into preferred ones", as Simon's (1996, p. 111) oft-quoted definition has it (see also Hunt, 2011; Otto and Smith, 2013). In this tension between the conventional orientations of anthropology and design, a key ambition in design anthropology is to develop a disciplinary exchange – or, as some argue, even a disciplinary hybrid – which constructively combines open-ended anthropological explorations of the present with designerly articulations of new possibilities in the future.

The field of design anthropology consists, however, of several distinct-yet-overlapping approaches and objectives. Overall, it is possible to identify four main types of relations between design and anthropology, as the former may constitute for the latter an object of critical inquiry (that is, anthropology *of* design), a field of practical engagement (that is, anthropology *for* design), a model for disciplinary rethinking (that is, anthropology *as* design), and an opportunity for interdisciplinarity (that is, anthropology *with* design) (Escobar, 2017; Murphy, 2016). Although these relations differ, it is a key point that their differences notwithstanding, design and anthropology converge around the human capacity to inquire into the possibilities of human life and, thus, to propose other or new ways of thinking, doing, and being.

In this way, a reorientation of design has emerged at the intersection of social theories and design studies, displacing the focus of design from material objects and solutions to its world-making capacities. As Escobar (2017, p. xvi, references omitted) so eloquently puts it: "Between 'the life of form' and 'the form of life' an entire design space opens up". Or, to quote Verbeek (2012, p. 163), "in practices of design [ … ] human beings do not only give shape to their material world but also to their own existence". In other words, design is essentially ontological in that it stakes out particular directions into the future that impact and shape ways of being. The notion of ontological design thus builds on this double movement; "namely, that we design our world, and our world designs us back – in short, design designs" (Escobar, 2017, p. 4). Importantly, it follows that the notion invites us to both recognise and develop how design may serve to reimagine and reconstruct local modes of living and being (Escobar, 2017). Anthropology feeds into this endeavour, we contend, not only because anthropologists by way of fieldwork seek to learn from and understand those local modes of living and being, but also because they, by so doing, speculate about the *possibilities* of life and, thus, "speak" new worlds into being, as Hastrup (1989) once put it.

While the notion of ontological design and the convergence of design and anthropology may perhaps seem a little grandiose for a teaching course

on a bachelor's programme, it nonetheless forms the backbone of our approach to case-based teaching. To recap, our aim was to train the students not merely in exploring questions and challenges posed by external partners but in rethinking and reconfiguring prevalent assumptions and actions – and, thus, in allowing for new possibilities or modes of being to emerge. We describe the specifics of our approach in the following.

## Applied anthropology and case-based teaching

As coordinator (Simon) and teachers (Simon and Kasper) of the mandatory course in applied anthropology, we have both been highly engaged in coordinating, teaching, facilitating, and evaluating all course activities, from early preparations to final exams. With the ambition of documenting the design anthropological, case-based teaching course, Simon has written ethnographic notes during the first two introduction weeks and the following 12 project weeks. During the latter, Simon moreover conducted a form of participatory observation by systematically following seven student projects involving 24 students, who all worked on issues related to circular economies and waste systems. Finally, the teaching group consisting of six anthropologists has held ongoing "touch-base meetings" during the course, which have also been described in written notes by Simon, emphasising shared discussions and reflections on core elements such as student projects, class structure, collaboration with external partners, extraordinary conditions forced by coronavirus lockdowns, and applied dimensions of anthropology as a discipline. In addition to these engagements, observations, and documentation, both of us have approximately ten years of experience with developing and teaching case-based courses in anthropology at Aarhus University and University of Copenhagen, which have guided not only our design anthropological approach to case-based teaching but also the argument in this chapter.

### *Course format and case challenge*

"Applied anthropology", a course on 15 ECTS, included all students on the fourth semester on the bachelor's programme in anthropology. The ambition of the course was to give students an opportunity to use their anthropological competences when exploring a concrete case challenge posed by an external organisation. In this sense, the students were to apply the methods, theories, and analytical approaches from their first three semesters with the purpose of delving into a contemporary societal challenge. The classes were taught by six anthropologists, of which four were internal scholars and two external professionals, which had been set up in this way in order to ensure that lectures, supervision, and facilitation could draw on experiences from both inside and outside academia. The learning process unfolded over 14 weeks with 28 teaching sessions of three hours, beginning

with two introductory weeks where all 69 students participated and formed project groups consisting of three to five people. The students were then divided into three relatively independent courses, each concerned with a specific case challenge posed by an external partner within the themes of food, health, and waste, respectively. These courses unfolded over 12 weeks in four stages: (1) case challenge and problem statement, (2) fieldwork, (3) analysis and ideation, and (4) dissemination.

In this chapter, our analysis and discussion lean primarily on the waste challenge, which was posed by the Municipality of Copenhagen. Entitled "From waste to resource", the case concerned themes such as waste management, recycling practices, and the circular economy. The ambition of the municipality was to understand, map, and upscale "good waste practices" performed by younger citizens in Copenhagen. In particular, they asked the students to explore the following questions: How do citizens experience their daily handling of waste? How can the municipality support recycling for different types of waste? What makes citizens experience something as a resource and not as waste? What does a good waste room contain? And how does the surrounding environment influence the citizens' handling of waste?

### *Preparation and coordination*

The teachers' preparation started during the autumn of 2020, months before the first class in the beginning of February 2021. The first task for Simon, as coordinator of the course, was to establish a team of six teachers. He reached out to two colleagues in his department, Kasper and Jane, who both had experience with teaching case-based and practice-oriented courses. Having accepted the task, the three teachers recruited a PhD student and two anthropologists working outside the university. Then, they partnered up with three organisations, which turned out to be a manageable task as anthropological competences and insights are commonly viewed as valuable by many public and private organisations in Denmark. The organisations were the Danish Alzheimer Association, the retail corporation Coop, and the Municipality of Copenhagen. In this process, it was crucial to ensure that employees at all organisations were willing to spend time on a range of activities such as, for instance, to participate in an introduction meeting with all students, to facilitate contact to interlocutors in the organisation, to take part in a status meeting halfway through the course, and to engage in a final three-hour seminar with student presentations.

Having reached an agreement with the organisations, two teachers then teamed up with one organisation with whom they formulated the case challenges. These challenges were presented to the students during the first introduction week in both written material and video presentations. Later, the same two teachers were responsible for developing and teaching the design anthropological stages (see below) linked to the case challenge. The

classes were divided into lectures introducing relevant case material, theoretical concepts, collaborative work in project groups, seminars across the project groups, and meetings with the organisations. Thus, the teachers took on the roles as lecturers and facilitators in that class teaching was continuously combined with project work. Before initiating classes, the teaching group held two coordination meetings in which Simon introduced the ambitions, structures, methods, and practicalities of the course. This was followed by ongoing "touch-base meetings" in which all teachers addressed experiences and issues of concern.

***Phase 0: Introduction (Weeks 1 and 2)***

All the students started with two introductory weeks of four teaching sessions. In this phase, the students were introduced to applied anthropology and design anthropology; they received a lecture on digital methods and they learned about collaborative work in short and intensive anthropological projects. After being introduced to the small videos and the written material from the partner organisations, all students sent an individual e-mail to Simon stating which of the three case challenges they would prefer to work on. Simon then divided the students, firstly, into the three overall sub-courses connected to each of the case challenges and partner organisations and, secondly, into project groups of three to five students.

***Phase 1: Problem statement and project design (Weeks 3 and 4)***

During the first week, the students met online with employees from each organisation. For the students collaborating with the Waste Unit in the Municipality of Copenhagen, the meeting focused on the case description, mutual expectations, fieldwork activities, and, more generally, the organisational strategies and procedures in the municipality. Although being in the middle of the coronavirus lockdown, the meeting on the digital platform Zoom provided the students with detailed information about the case challenge, which they could use in the formation of their projects. The meeting thus served both as a data source and as an opportunity to get access to other interlocutors in the municipality. It was, in other words, the outset for the coming project weeks as the students reflected critically on the case description, developed an anthropologically informed problem statement, designed fieldwork activities, and took initial contact to interlocutors in the Copenhagen area. In parallel, they spent time on aligning how they were internally to collaborate during the next 12 weeks. As an output, the students handed in their first portfolio assignment (five pages), containing a problem statement and related research questions, as well as a project design outlining methods and activities, work division and roles, and considerations about possible ethical implications and data management.

***Phases 2 and 3: Pilot studies and fieldwork (Weeks 5–8)***

During the first class in the second phase, each group of students had a 20-minute status meeting with the two teachers. The students received feedback on their written portfolio assignment, followed by a dialogue about the upcoming activities. In parallel, the other groups prepared their fieldwork activities and secured access to relevant field sites in Copenhagen. Then, during weeks six, seven, and eight, the students conducted their fieldwork, consisting of semi-structured interviews with relevant citizens, participatory observation in outdoor waste environments, informal talks with people in public spaces, quantitative surveys on digital platforms, and focus group interviews with professionals in primarily the Waste Unit. The coronavirus lockdown continued throughout the pilot and fieldwork phases, which meant that the students also developed online tools such as, for instance, digital diaries for interlocutors to describe their waste practices in pictures and instant messages. They used video calls on smartphones to digitally "shadow" people when performing waste practices in their private homes or when using outdoor waste spaces. Although conducting ethnographic fieldwork proved somewhat difficult in an extraordinary situation of physical distancing, all groups managed to develop a solid data foundation for their analysis and ideation. The empirical data and reflections on the project progress were finally described in the second written portfolio assignment (five pages).

***Phase 4: Analysis and ideas for actions (Weeks 9–11)***

After their fieldwork and the feedback from teachers on their second portfolio, as well as after a halfway meeting with the gatekeeper in the municipality, the students began to analyse their empirical material. To this end, the teachers had presented online video lectures during the course, introducing both theoretical concepts from anthropology and other disciplines and empirical examples concerning themes such as circular economy, waste systems, sustainable communities, green living experiments, and organisational change. The idea was to inspire the students to pursue new insights and ideas relevant to the problem statement, case challenge, and disciplinary background. By reaching new understandings of, and perspectives through, their analytical work, the aim was to build a solid platform for new ideas and designs for actions to emerge. Having worked with mostly practical components such as the case challenge, fieldwork, and empirical data, the students found it challenging to shift to a more analytical mode. Yet, as we shall elaborate on in our analysis and discussion, the students managed to create analytical insights and develop them into design proposals for new initiatives and actions to be taken in the municipality. This was the focus of the third written portfolio assignment (seven pages).

***Phase 5: Ideation, dissemination, and project reflections (Weeks 11–14)***

Having once again received feedback on their written portfolio assignment, the students were now tasked with creating visual presentations of their key findings. To this end, an external anthropologist working in a consultancy in Copenhagen shared his insights on how to develop compelling, reliable, and practice-oriented "pitches". The students turned their descriptions and analysis into oral presentations of 12 minutes (the fourth portfolio), which they then presented to the employees in the Waste Unit during a three-hour Zoom seminar in week 13. Importantly, each of the 12-minute presentations was followed by a 10-minute dialogue between the employees and the students. After the presentations, the students finalised the course by composing individual reflections (the fifth portfolio) on their design anthropological work, concerning such themes as digital ethnography, collaboration in projects, and application of anthropology to problems posed by external organisations.

***Post-course: The exams***

Two weeks after the final class, the students handed in three deliveries as their exam: (1) all their written portfolio assignments (project design, fieldwork report, analysis, and reflections) in revised form, (2) a video recording of their oral presentations (the fourth portfolio), and (3) a short visual presentation of their key insights and conclusions. The exam, thus, contained both written and oral deliveries that were to be assessed by the teachers and two external examiners. Three weeks after the deadline for handing in the exam, the teachers and the external examiners met online to assess and grade the exams, followed by an online opportunity for the students to receive verbal feedback on their overall performance.

## Case-based teaching in action and "careful littering"

Having outlined the structure and progression of our approach to case-based teaching, inspired by the attempt in design anthropology to combine open-ended anthropological explorations and novel design proposals, we now turn to a concrete student project on "careful littering" in order to more clearly elucidate the impact achieved in and through the design anthropological learning process.

"We have decided to focus on people's careful handling of waste", Johanne explains at our first status meeting in the second week of the course. "The meeting with the municipal actors made us reflect on the values of handling waste in the city, … [and] we want to examine how public spaces can be turned into environments for careful waste handling". While curiously following our response, Johanne elaborates that the group

will investigate the relation between humans and waste through the concept of care. Ellen and Siri, who are part of the group, add that the project concerns the fact that the municipality is commissioned to set up 750 new sorting points in public spaces in Copenhagen. "We think that it is interesting to see if waste handling in the city could go hand in hand with a sense of care for the environment", they say. Ellen looks at Halfdan, another member of the group, who explains that they understand waste handling as influenced by both social values and material infrastructures. Halfdan concludes: "You should treat it [waste] as something useful, you know, as we talked about in the meeting with the municipality … Waste as a resource … We need to take better care of our resources".

To explore the idea of careful handling of waste, the group conducted fieldwork over a period of four weeks, including semi-structured interviews, focus group discussions, and participant observation in public spaces in Copenhagen. In interviews with citizens living in Copenhagen, several interlocutors explained that they defined and handled waste in streets and parks quite differently from their private homes. A young woman, for instance, said:

> It is as if it is very different in private homes. There, you care about waste and you do it. But in public spaces, you don't do anything. It is like different social rules apply here.

With this insight in mind, the students initiated their participant observation by going into the streets of Copenhagen, with the ambition of reaching an understanding of the "social rules" of waste practices in public spaces. During their fieldwork, they repeatedly observed how people did not seem to notice the waste lying around in streets and parks. The students therefore suggested that waste had become a natural component of the public space, meaning that people were simply not conscious about litter in streets and parks. This, the students argued, was a key reason for why people, although perceiving waste as an important environmental concern, did not do anything about it such as, for instance, picking up the litter right in front of them. These observations led the students towards the question of how they could, so to speak, open people's eyes to all the waste scattered in public spaces, and thereby change their attention and practices towards it.

In parallel, the students also participated in two waste collecting events, organised by the Danish Society for Nature Conservation and an NGO called *Skraldefest* (literally translating into "Waste Party"). During the events, the students talked to participants who perceived the act of picking up waste on various scales; for them, it represented simultaneously a protective and careful concern for parks and streets in the city and an opportunity for participating in actions linked to a broader green transition of society. The students thus took note of how the participants aligned a more general concern and care for the planet with a specific ambition to clean up

the city through waste collection practices. Furthermore, they experienced recognition and gratitude from people they randomly met during their waste collection; for instance, a passing jogger who declared his support by shouting "good work", while others shared an encouraging "well done". Likewise, when allowing a young man to throw his litter into their sacks, he folded his hands and bowed, followed by an appreciative "thank you". The students thus became aware of how the participants, including themselves, turned not only their own but also other people's attention towards the issue of waste collection in public spaces. With a black waste sack in their hands, they now detected all the waste that they had not laid their eyes on before. Importantly, this "awakening" did not lead to an increased concern as it, in a sense, happened in the act of finding a solution to the very same problem. The students explained that, after the events, waste in public spaces still seemed to "catch their eyes", which they encapsulated in a quote from one of the organisers: "once seen, it can't be unseen". Finally, the students concluded that "awakening events", such as waste collections, could open people's eyes and give them an incentive to do something; not just about their own waste, but also about litter discarded in streets and parks by others.

Framing this as "careful littering", the students presented these ethnographic insights to employees in the municipality, recommending them to design and introduce awakening events and infrastructures which would engender a re-objectifying of unnoticed waste. Besides proposing that the municipality should thus support events that engage citizens in public waste collections, they furthermore outlined an idea, by way of visual sketches (see Figure 9.1), of making existing waste options more visible through unconventional designs and colours, thereby materially reminding people of the importance of throwing out waste in garbage cans. In this way, the

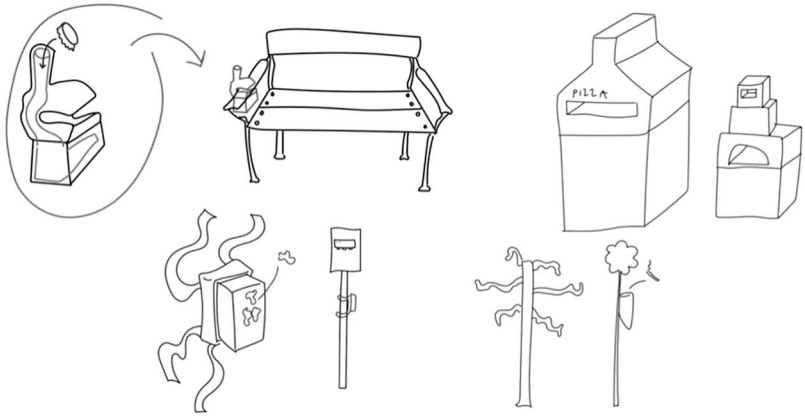



*Figure 9.1* Sketches of unconventional garbage cans developed by the group.

students combined anthropological insights and designerly proposals in their recommendation to facilitate both social events and material infrastructures that would re-objectify waste and waste handling practices – and, by implication, generate new modes of "seeing" and "doing" waste in public spaces in Copenhagen.

## Collaborative tensions and anthropological impact

While the above description of the student project may appear rather frictionless, it was not, of course. Tensions and challenges arose continuously, particularly because of the often differing ideas and purposes of a university and a municipality, respectively. As such, our case may be seen as an empirical window into some of the complexities and intricacies of the encounter between higher education and public organisations – and, more broadly, between humanities and business (broadly understood). In this respect, it should be noted that the collaboration involved no money or business contract, which may seem as a minor detail, but which has major importance in terms of responsibility and authority. The conditions framing the collaboration were based on mutual interests and potential gains for both parties, meaning that students as well as municipal actors were free to influence the projects in certain directions while also having to take account of each others' concerns in order for the collaboration to be of value. Negotiations and alignments, in other words, were an intrinsic part of the process, not least when tensions arose. In what follows, we discuss the most significant tensions which revolved around the question of making an impact.

From the perspective of the municipality, an idea of knowledge production in higher education as mainly driven by internal disciplinary discussions and theories soon caused Phil, our collaborating partner in the Waste Unit, to worry. In particular, he expressed a concern about what, "in his world", would be seen as too many abstract ideas and a lack of focus on applicable solutions. The students, on the other hand, worried rather about the opposite: that the municipal focus on solutions would mean that their projects would develop in predetermined directions, thus compromising the open, exploratory nature of anthropological research. For Simon as teacher, it was, therefore, pivotal to explain and insist on the anthropological approach in meetings with Phil, while also encouraging the students to see the expectations of the municipality as empirical data which could help them to understand the organisational dynamics and value criteria in the organisation. The encounter between the differing approaches and objectives of university students and municipal actors thus required the teacher to take on a mediating and aligning role.

For the students, tensions moreover emerged in relation to the course requirement of translating their anthropological insights into a practice-oriented oral presentation with design implications. Faced with the

challenge of developing a vocabulary suitable for the work of the municipal actors, some of the students found the dissemination to be a "reduction" of their anthropological knowledge, as it necessarily had to leave out certain questions, insights, theories, and other considerations. Although insisting on training the students' abilities to disseminate their work in ways that was understandable and meaningful to municipal actors, the teachers stressed that the presentations were not to be viewed as final solutions. Rather, they should serve as "invitations" or "openings" to constructive dialogues, being not operational solutions, but novel perspectives that might inspire the municipality to rethink and redesign existing practices and systems. By understanding the students' insights and results as a kind of prototype – that is, as indefinite and open-ended (Jiménez, 2014) – the collaboration thus progressed as a joint exploration of both present and future waste handling in the municipality and beyond.

In fact, in the concrete encounter between an insistence on academic virtues and an urge for practical usefulness, what seemed to provoke interest and curiosity, as well as new questions, among the municipal employees, was particularly the invitation to understand and approach waste systems and practices in hitherto unprecedented ways. Immediately after a student presentation, for instance, an employee responded with excitement:

> One of your first insights was really interesting; the one about the black box where you identified pleasure as an aspect that stimulates waste handling. I would like to hear more about that … It's something that we have never thought about before, and it would be exciting to see if we could explore this aspect of pleasure even further.

A student from the group responded that the aspect of pleasure was closely linked to an overall ambition of making the world a greener place to live. She added that, although being dirty and smelling bad, the act of collecting waste in private homes, and seeing it accumulate in the waste bin, was broadly perceived as satisfying because it represented an answer to a general concern about pollution and climate changes. Demonstrating a pleasure in collecting litter and seeing waste handling as an act of environmental care thus induced a creative and "useful" exchange between the students and the employees. Later, Phil enthusiastically wrote to Simon that the ideas and concepts developed by the students were now articulated and referred to by his colleagues in their everyday work with designing new ways of handling waste in Copenhagen.

As such, what triggered excitement and proved to be of value was not fixed solutions to pre-defined questions posed by the Waste Unit. Rather, the students reflected critically on the case challenge, developed their own problem statement, explored it by way of fieldwork, analysed the empirical material, and presented insights and recommendations which, like prototypes, did not confirm, but opened up commonplace understandings and

practices. By bringing anthropological explorations of *present* circumstances and designerly articulations of *future* possibilities into dialogue, the students thus proposed new perspectives on how to perceive, act, and be in a world with increasing amounts of waste. The unfinished, prototypical, and fluid nature of their proposals engendered a creative space, not only for jointly discussing them, but also for adjusting and modelling them into existing systems, projects, and ideas in the municipality. Being neither fixed solutions to practical problems (as often associated with "business") nor merely critical reflections on a complex issue (as often associated with "humanities"), the ambition was to foster and encourage a critical approach of reconfiguring extant modes of thinking, doing, and being – what we, following Escobar (2017), have termed *ontological design* – which would have significant impact on waste handling. In a nutshell, what emerged was critical thinking and constructive application as two sides of the same coin.

## Conclusion

In this chapter, we have provided a hands-on description of a distinct approach to case-based teaching as a way to organise practice-oriented courses in anthropology and allied disciplines. While positioned within a broader context of contemporary science policies, which highlight open collaboration, mission-oriented agendas, and societal impact, we have argued for an approach that encourages students to critically yet constructively deconstruct and reconfigure prevailing understandings and practices. More specifically, the students are urged to bring together open-ended anthropological explorations and design proposals for new modes of acting and being in the world. Approaching the students' insights and recommendations not as tools and solutions, but as fluid and open prototypes allows for the students and their collaborating partners to jointly explore and discuss novel perspectives and procedures, as well as to integrate them into existing systems, projects, and ideas. The students thus adopt a critical yet constructive approach by way of reconfiguring prevailing ways of thinking, doing, and being, which may lead to innovative impacts in and for society.

In response to the urgent need for new perspectives on current and future sustainable societies, we contend that such an approach is crucial to challenge and rethink existing orders, logics, and procedures. Unlike political actors, public organisations, and private corporations, academic researchers and university students are in a particular "fringed" position for proposing and designing ontological reconfigurations in concrete domains of contemporary societies. This means that collaborating with public and private actors in an exploration of societal challenges or missions may engender a learning process in which academic virtues and ethics – such as empirical commitment, rigorous analysis, and critical reflection – are put to use and have impact in collaborative, creative, and societally engaging ways.

Finally, this links up to a significant question posed by Shore and McLauchlan (2012, p. 282); that is, "what kinds of subject or 'citizenship' are being promoted by university entrepreneurialism?" In our approach, we believe that case-based teaching essentially trains the students in openly and critically exploring key societal questions with the aim of contributing actively and constructively to them. As probably any form of education, it has subject effects in the sense that education, as Ingold (2018) argues, is not about transmission of authorised knowledge but a way of attending to things. Following this argument, our hope is to inspire the students to explore and rehearse how they may act as critical yet engaged citizens who share and perform their anthropological insights and ideas to a variety of actors in society, with the purpose of proposing and designing new sustainable ways of being in the world.

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## References

Burke, M. G., Carter, J. D. and Hughey, A. W. (2013). The use of case study competitions to prepare students for the world of work. *Industry and Higher Education*, *27*(3), 157–162.

Damsholt, T. and Jespersen, A. P. (2014). Innovation, resistance or tinkering: Rearticulating everyday life in an ethnological perspective'. *Ethnologia Europaea*, *44*(2), 17–30.

Escobar, A. (2017). *Designs for the Pluriverse: Radical Interdependence, Autonomy, and the Making of Worlds*. Durham & London: Duke University Press.

Etzkowitz, H. (2008). *The Triple Helix: University-Industry-Government Innovation In Action*. New York & London: Routledge.

Etzkowitz, H. and Leydesdorff, L. (eds.) (1997). *Universities in the Global Economy: A Triple Helix of University-Industry-Government Relations*. London: Cassell Academic.

Gibbons, M., Limoges, C., Nowotny, H., Schwartzman, S., Scott, P. and Trow, M. (1994). *The New Production of Knowledge: The Dynamics of Science and Research in Contemporary Societies*. London: Sage Publications.

Gibson-Graham, J. K. (2008). Diverse economies: Performative practices for "other worlds". *Progress in Human Geography*, *32*(5), 613–632.

Gunn, W. and Donovan, J. (eds.) (2012). *Design and Anthropology*. Surrey, England: Ashgate.

Gunn, W., Otto, T. and Smith, R. C. (eds.) (2013). *Design Anthropology: Theory and Practice*. London & New York: Bloomsbury.

Halse, J. (2013). Ethnographies of the Possible. In Gunn, W., Otto, T. and Smith, R. C. (eds.), *Design Anthropology. Theory and Practice*. London & New York: Bloomsbury, pp. 180–196.

Hastrup, K. (1989). The Prophetic condition. In Chapman, M. (ed.), *Edwin Ardener: The Voice of Prophecy and other essays*. Oxford: Basil Blackwell, pp. 224–228.

Hunt, J. (2011). Prototyping the social: Temporality and speculative futures at the intersection of design and culture. In Clarke, A. J. (ed.), *Design Anthropology: Object Culture in the 21st Century*. Wien & New York: Springer, pp. 33–44.

Ingold, T. (2018). *Anthropology and/as Education*. Abingdon, Oxon & New York: Routledge.

Jensen, T. E. (2013). Techno anthropology – a new move in science and technology studies'. *STS Encounters*, *5*(1), 1–22.

Jespersen, A. P., Petersen, M. K., Ren, C. and Sandberg, M. (2012). Guest editorial: Cultural analysis as intervention. *Science Studies*, *25*(1), 3–12.

Jiménez, A. C. (2014). Introduction. The prototype: More than many and less than one. *Journal of Cultural Economy*, *7*(4), 381–398.

Krause-Jensen, J. and Garsten, C. (2014). Introduction: Neoliberal turns in higher education. *Learning and Teaching*, *7*(3), 1–13.

Loi M. and Guardo, M. C. D. (2015). The third mission of universities: An investigation of the espoused values. *Science and Public Policy*, *42*(6), 855–870.

Mazzucato, M. (2021). *Mission Economy: A Moonshot Guide to Changing Capitalism*. Dublin: Penguin Books.

Mazzucato, M. (2018). *Mission-Oriented Research & Innovation in the European Union: A Problem-Solving Approach to Fuel Innovation-Led Growth*. Brussels: European Commission.

Moallem, M., Hung, W. and Dabbagh, N. (eds.) (2019). *The Wiley Handbook of Problem-Based Learning*. Hoboken, NJ: Wiley Blackwell.

Murphy, K. M. (2016). Design and anthropology. *Annual Review of Anthropology*, *45*, 433–449.

Nowotny, H., Scott, P. and Gibbons, M. (2001). *Re-Thinking Science: Knowledge and the Public in an Age of Uncertainty*. Cambridge & Maiden, MA: Blackwell.

Otto, T. and Smith, R. C. (2013). Design anthropology: A distinct style of knowing. In Gunn, W., Otto, T. and Smith, R. C. (eds.), *Design Anthropology: Theory and Practice*. London & New York: Bloomsbury, pp. 1–29.

Ranga, M. and Etzkowitz, H. (2013). Triple helix Systems: An analytical framework for innovation policy and practice in the knowledge society. *Industry & Higher Education*, *27*(3), 237–262.

Shore, C. and McLauchlan, L. (2012). "Third Mission" activities, commercialisation and academic entrepreneurs. *Social Anthropology*, *20*(3), 267–286.

Shore, C. and Wright, S. (2017). Introduction: Privatizing the public university: key trends, countertrends and alternatives. In Wright, S. and Shore, C. (eds.), *Death of the Public University? Uncertain Futures for Higher Education in the Knowledge Economy*. New York & Oxford: Berghahn, pp. 1–27.

Simon, H. A. (1996). *The Sciences of the Artificial*. Cambridge, MA: The MIT Press.

Smith, R. C., Vangkilde, K. T., Kjærsgaard, M. G., Otto, T., Halse, J. and Binder, T. (eds.) (2016). *Design Anthropological Futures*. London & New York: Bloomsbury.

Verbeek, P.-P. (2012). Introduction. humanity in design: A few notes on the relations between anthropology and materiality. In Gunn, W. and Donovan, J. (eds.), *Design and Anthropology*. Surrey, England: Ashgate, pp. 163–176.

Whelan, A., Walker, R. and Moore, C. (eds.) (2013). *Zombies in the Academy: Living Death in Higher Education*. Chicago: The University of Chicago Press.

Wright, S. and Shore, C. (eds.) (2017). *Death of the Public University? Uncertain Futures for Higher Education in the Knowledge Economy*. New York & Oxford: Berghahn.

Zomer, A. and Benneworth, P. (2011). The rise of the university's third mission. In Enders, J., Boer, H. F. d. and Westerheijden, D. F. (eds.), *Reform of Higher Education in Europe*. Rotterdam: Sense Publishers, pp. 81–101.